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LARSON—MATH 756—HOMEWORK WORKSHEET 04

A *vertex cover* in a graph is a set of vertices that is incident to each edge of the graph. The *covering number* β is the minimum cardinality of any vertex cover.

A vertex v of a graph is *simplicial* if its closed neighborhood $N[v]$ is a clique.

1. Show that if v is a simplicial vertex then there is a maximum independent set containing v . (Simplicial vertices can be identified efficiently) so this is one way to reduce the maximum independent set (MIS) problem to a smaller graph, namely $G - N[v]$.

König's Theorem (or the König-Egervary Theorem). For a bipartite graph, $\alpha + \mu = n$, where μ is the matching number.

König's Theorem (Mini-max version) . For a bipartite graph, $\mu = \beta$ (where β is the cardinality of a minimum vertex cover, that is, a minimum cardinality set of vertices incident to every edge in the graph).

2. Show that the two versions of König's Theorem are equivalent.
3. Let G be a connected graph and $v \in V(G)$. Let $E = \{w : d(v, w) \text{ is even} \}$ and $O = \{w : d(v, w) \text{ is odd} \}$. Show: G is bipartite if and only if E and O are independent sets.
4. Let the **bipartite number** $b = b(G)$ of a graph G be the cardinality of a maximum cardinality induced bipartite subgraph. So $\alpha(G) \geq \frac{b}{2}$. Find a class of non-bipartite graphs where $\alpha = \frac{b}{2}$.
5. Find a class of graphs where $\alpha - \frac{b}{2}$ is not bounded. (That is, find a collection of graphs $\mathcal{G}$, where for any k , there is a graph G in $\mathcal{G}$ with $\alpha(G) - \frac{b(G)}{2} \geq k$).